Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/31/2007
PI Name:	Juergensmeyer, Margaret A Ph.D.		
Project Title:	Gene Expression Profiles for E. coli in Common Modeled Microgravity Systems		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mjuergensmeyer@IITRI.org	**Fax:**	FY 312-567-4852
PI Organization Type:	NON-PROFIT	**Phone:**	312-567-4918
Organization Name:	IIT Research Institute		
PI Address 1:	Microbiology and Immunology Division		
PI Address 2:	10 W. 35th St.		
PI Web Page:			
City:	Chicago	**State:**	IL
Zip Code:	60616	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	05/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	2/07--Received NCE to 5/31/2007, per A. Chu (ARC). 8/06--funding reinstated thru 2/28/07, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Juergensmeyer, Elizabeth (Judson College) Mobley, Evelyn (IIT Research Institute)		
Grant/Contract No.:	NNA04CC75G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bacteria are known to respond to the spaceflight environment. Literature reviews reveal a wide array of responses, from antibiotic resistance to changes in growth curve kinetics. However, because it is extremely rare for an investigator to perform experiments on multiple flights, and because flight hardware is not standardized, it is often difficult to determine which element of the spaceflight environment is responsible for the observed physiological change. It is often assumed that microgravity or microgravity-induced physical changes (e.g. lack of convective movement) are responsible, but at present, there is no way of confirming this. In addition, for nearly every paper that identifies a change and attributes that change to the spaceflight environment, there is a paper that mentions no change observed. This lack of change may be even more prevalent than the literature reveals, as it is difficult to publish a paper that shows no changes in response to the treatment. Bacteriologists interested in the response of their chosen organism to the spaceflight environment often utilize a variety of modeled microgravity (MMG) systems. Slow clinostats (approx. 1/4 rpm), fast clinostats (approx. 60 rpm), and High Aspect Rotating Vessels (HARVs) have all been used as MMG systems. Previous research from this laboratory has determined that linear vibration, a facet of the spaceflight environment, can induce some of the physiological changes observed during and attributed to spaceflight itself. The primary goal of this project was to identify the changes in gene expression that occur in various types of MMG, with the future goal of comparing them with changes shown during actual space flight. We proposed to meet this primary goal by growing Escherichia coli K-12, a common research organism that has often flown on orbital platforms, in a wide array of MMG systems. Growth curves would be performed in each MMG, and bacterial samples would be collected at equivalent points in the growth curve. Total RNA would be isolated from the samples and analyzed using Affymetrix gene chip microarrays, which would identify changes in gene expression in each sample. A comparison across the MMGs would firstly identify genes that were changed in response to the known MMGs, thus indicating which pathways were activated in a given ground control or experimental treatment. In addition, a future flight experiment, whether done by ourselves or another researcher, can be compared to this data, allowing researchers to identify which MMG is most similar to flight results. This will allow the formation of baseline data, so that at some point in the future, flight cultures can be examined utilizing the same techniques, and a comparison of the flight cultures and the baseline data will allow identification of the optimal MMG that mimics the spaceflight environment. In addition, a comparison of MMGs will identify those that best model specific subsets of the microgravity system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to ultimately enabling the identification of an optimal MMG for future ground research, this project is expected to identify gene pathways connected with motility, enhanced growth, and nutrient utilization in E. coli.
Task Progress:	In the last year of this project, analysis of the microarray data has continued. A poster on this data was presented at the American Society for Gravitational and Space Biology annual meeting in November, 2006. Further research correlating the microarray data with physiology associated with space flight was conducted. Changes in antibiotic susceptibility, including increases and decreases in susceptibility, have been observed during and immediately post-spaceflight. E. coli were grown in the four modeled microgravity (MMG) systems, and at mid-log phase were exposed to a suite of up to 11 different antibiotics. Minimal Inhibitory Concentrations were determined for each antibiotic at each growth condition. A correlation with the microarray results indicates that changes in antibiotic resistance approximately equivalent to those observed during and after spaceflight were found on all MMGs. However, slow clinorotation, which had the highest number of upregulated antibiotic resistance genes, showed the least changes in actual resistance. This may indicate a delay of onset of gene upregulation in this MMG.
Bibliography Type:	Description: (Last Updated: 09/04/2007)
Abstracts for Journals and Proceedings	Juergensmeyer EA, Juergensmeyer MA, Mobley EM. "Differences in gene expression in E. coli grown on four common modeled microgravity systems." Presented at the 22nd Annual Meeing of the American Society for Gravitational and Space Biology, Arlington, VA, November 2-5, 2006. Gravitational and Space Biology. 2006 Nov;20(1):22. , Nov-2006